

Sonya Simkin

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Education

Radboud University , Software Science	Nijmegen, Netherlands Jan 2026 – Jan 2030
Carnegie Mellon University , Computer Science	Pittsburgh, Pennsylvania, USA Jan 2024 – Jan 2025
Carnegie Mellon University , Computer Science	Pittsburgh, Pennsylvania, USA Jan 2020 – Jan 2024

Experience

CMU CS Academy , Software Engineering Intern - Developed Rust-based library for Python graphics framework in Desktop CMU Graphics using Skia for graphics rendering and winit for windowing - Implemented frontend changes in React and Redux for the CS Academy online learning platform, including improving search result display and site option accessibility - Improved backend data collection using the Django framework in Python	Pittsburgh, PA, USA Oct 2025 – Aug 2026 11 months
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Awards

Mark J. Stehlik Introductory and Service Teaching Award The Stehlik Award acknowledges those bringing their best and indispensable efforts to the task of introductory teaching. The selection is based on student nominations, recommendation letters, and reviews, and is evaluated by the SCS Teaching Excellence Committee. Sonya Simkin www.cs.cmu.edu/~scsfacts/stehlikaward.html	Jan 2026
Senior Leadership Recognition Senior Leadership Recognition honors graduating seniors who have made a significant impact on our community and provides an opportunity for these outstanding students to reflect on their leadership at Carnegie Mellon and apply the lessons learned to their future goals and aspirations. Sonya Simkin www.cmu.edu/student-affairs/celd/leadership-development/senior-leadership/index.html	Jan 2024

Publications

Logical Relations for Noninterference with Cryptography Master's thesis Sonya Simkin reports-archive.adm.cs.cmu.edu/anon/anon/home/ftp/usr0/ftp/2025/CMU-CS-25-145.pdf	
TBD Pedestrian Data Collection: Towards Rich, Portable, and Large-Scale Natural Pedestrian Data Allan Wang, Daisuke Sato, Yasser Corzo, Sonya Simkin, Abhijat Biswas, Aaron Steinfeld arxiv.org/abs/2309.17187	

Manuscripts

Mechanizing a Proof-Relevant Logical Relation for Timed Message-Passing Protocols

Tesla Zhang, Asher Kornfeld, Rui Li, Sonya Simkin, Yue Yao, Stephanie Balzer

arxiv.org/pdf/2511.19521

A Language-Agnostic Logical Relation for Message-Passing Protocols

Tesla Zhang, Sonya Simkin, Rui Li, Yue Yao, Stephanie Balzer

arxiv.org/pdf/2506.10026

Languages

English

Native speaker

Russian

Heritage speaker

Projects

GradescopeOCaml

Jan 2026

An implementation of the [SavannahProject](#) autograder infrastructure for OCaml

CompileBPV

Jan 2024

A compiler for a small functional language using call-by-push-value (CBPV) as an intermediate representation, based on a talk by Max S. New